

API Development & Integration

RESTful Architecture, GraphQL, Webhooks, gRPC & Distributed Systems

REST & GRAPHQL ARCHITECTURE

1. Richardson Maturity Model & Level 3 HATEOAS

Design hypermedia-driven APIs returning dynamic transition links based on resource state.

2. Distributed Idempotency Framework

Intercept Idempotency-Key headers, lock in-flight requests, and cache response payloads with Redis.

3. Solving N+1 in GraphQL with DataLoader

Batch and cache database queries across concurrent resolvers to eliminate N+1 latency.

4. Rate Limiting via Sliding Window Counter

Implement Redis Lua scripts enforcing per-minute request quotas with standard retry headers.

WEBHOOKS & HIGH-PERFORMANCE RPC

5. Cryptographic Webhook Signatures (HMAC-SHA256)

Sign payloads with shared secrets, embed timestamp headers, and verify signatures with constant-time equality.

6. Webhook Fan-Out & Dead Letter Queue (DLQ)

Handle 100k events/min with event brokers, auto-scaling workers, and automated DLQ retry policies.

7. High-Performance gRPC Microservices

Implement Protocol Buffers proto3, bidirectional streaming, and interceptor-based tracing.